

U1-D3 Homework – Factoring Quadratics**Name:**

Simplify completely.

1. $x \cdot x^2 \cdot x^4$

2. $-3x^3(2x^2 - 3x + 5)$

3. $(2x - 11)(x - 4)$

Factor out a GCF.

4. $27x^8 - 18x^2$

5. $2x^6 + 32x^3 - 4x^2$

Factor completely. Some of these also have GCF's – pull them out first!

6. $x^2 + 18x + 72$

7. $2x^2 - 12x + 10$

8. $x^2 - 7x - 30$

9. $7x^2 - 15x + 2$

10. $3x^2 + 16x + 16$

11. $5x^2 - 7x - 6$

12. $4x^2 + 4x - 120$

13. $8x^2 + 51x + 18$

Solve.

14. $2x - 7 = 0$

15. $(3x - 4)(2x - 9) = 0$

Solve by factoring.

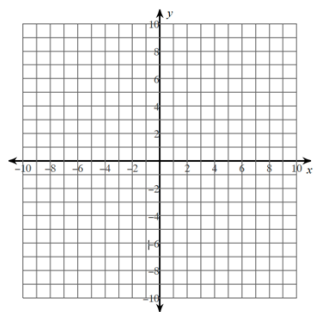
16. $x^2 - 28 = -12x$

17. $3x^2 - 20x + 32 = 0$

18. Find an equation in slope-intercept form for a line passing through $(6, -3)$ and $(-2, 1)$.

19. Solve by graphing.

$$\begin{aligned} 3 &= 3y - x \\ -24 + 6y &= -4x \end{aligned}$$



20. Solve by substitution.

$$\begin{aligned} x - 2y &= 13 \\ -3x - y &= -4 \end{aligned}$$